

GSFC NEON Series-1 CMO

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Near Earth Orbit Network (NEON) Series-1 Project, Code 473

**Near Earth Orbit Network
Series-1 Project Sounder for Microwave-
Based Applications
Instrument Mission Assurance
Requirements**



**Goddard Space Flight Center
Greenbelt, Maryland**

Near Earth Orbit Network Series-1 Project Sounder for Microwave-Based Applications Instrument Mission Assurance Requirements

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Preface

This document is under NEON Series-1 (NS1) Project configuration control. Once this document is approved, NS1 approved changes are handled in accordance with Class I and Class II change control requirements as described in the Low Earth Orbit (LEO) Programs Division (LPD) Programs and Projects Configuration Management Procedure (CMP), and changes to this document shall be made by complete revision.

Any questions should be addressed to:

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Change History Log

Revision	Effective Date	Description of Changes
Revision –	February 7, 2023	This version incorporates 470-CCR-22-0371 which was approved by the SMBA ERB on December 22, 2022, and by the SMBA CCB on the effective date shown.
Revision A	December 16, 2024	This version incorporates 470-CCR-24-0403 which was approved by the SMBA ERB on March 26, 2024, and incorporates NS1-CCR-0002 by the NS1 SMBA CCB on the effective date shown. Update the phase A IMAR to incorporate changes to the Code 300 class C baseline MAR, dated 6/20/2024. There are editorial and requirement clarification changes. The Document number has changed from 473-SMBA-RQMT-00005 to NS1-SMA-REQ-0002 as the document has transferred from the Management Information System (MIS) to the Technical Data Management System (TDMS).
Revision B	April 1, 2025	This version incorporates NS1-CCR-0006 which was approved by the NS1 SMBA CCB on the effective date shown. This version incorporates changes to the DID format to align with SMBA CDRL and changes DID MA 1-1.

Table of Deviations/Waivers

Item No.	Deviation/Waiver	Waiver Approved	Effectivity

Table of TBDs/TBRs/TBSs

Item No.	Location	Summary	Individual/ Organization	Due Date

Table of Contents

A. SCOPE, ORGANIZATION, AND CONVENTIONS.....	1
1 GENERAL.....	3
1.1 Systems Safety and Mission Assurance Program.....	3
1.2 Management.....	3
1.3 Requirements Flow Down	3
1.4 Identification of Project-Level Critical Items	3
1.5 Suspension of Work Activities	4
1.6 Surveillance.....	4
1.7 Government Mandatory Inspection Points	4
1.8 List of Sub-Tier Suppliers.....	5
1.9 Use of Inherited Products/Items.....	5
1.10 Protection of Flight Hardware	5
1.11 Risk Management	5
2 QUALITY MANAGEMENT SYSTEM	6
2.1 General.....	6
2.2 Supplemental Quality Management System Requirements.....	6
2.2.1 Control of Nonconforming Product	6
2.2.2 Material Review Board	6
2.2.3 Anomaly Reporting and Disposition	7
2.3 Orbital Debris Assessment Report and End of Mission Plan	7
3 SYSTEM SAFETY.....	8
3.1 General.....	8
3.2 Mission Related Safety Requirements Documentation	8
3.3 System Safety Deliverables	9
3.3.1 System Safety Plan.....	9
3.3.2 Safety Requirements Compliance Checklist	9
3.3.3 Instrument Safety Assessment Report.....	9
3.3.4 Hazard Analyses.....	9
3.3.4.1 Preliminary Hazard Analysis	9
3.3.4.2 Operations Hazard Analysis and Hazard Verification Tracking Log.....	10
3.3.4.3 Operating and Support Hazard Analysis	10
3.3.5 Verification Tracking Log.....	10
3.3.6 Hazardous Procedures for Payload I&T and Pre-launch Processing	11
3.3.6.1 Lifting Device Safety Requirements	11
3.3.7 Mishap Reporting and Investigation	12
3.3.8 NASA Payload Safety Program Forms	12
4 RELIABILITY.....	13
4.1 Reliability Program Plan.....	13
4.2 Analysis of Design.....	13
4.2.1 Failure Modes and Effects Criticality Analyses and Critical Items List.....	13
4.2.2 Fault Tree Analysis	16
4.2.3 Reliability Calculations	16
4.3 Limited Life Analysis	16

4.4	Parts Stress Analysis	17
4.5	Worst-Case Analysis.....	17
5	SOFTWARE ASSURANCE	18
5.1	Software Assurance Program.....	18
5.2	Surveillance of Software Development, Maintenance, and Assurance Activities.....	18
6	WORKMANSHIP	20
6.1	General.....	20
6.2	Electrostatic Discharge Control	20
6.3	Printed Circuit Boards.....	20
6.4	Lead-Free Control Measures.....	21
7	ELECTRICAL, ELECTRONIC, AND ELECTROMECHANICAL (EEE) PARTS	22
7.1	General.....	22
7.2	Parts Control Board.....	22
7.3	Re-use of EEE Parts.....	22
7.4	Master EEE Parts List.....	22
7.5	Radiation Effects Mitigation.....	22
8	MATERIALS AND PROCESSES	23
8.1	M&P Selection, Control, and Implementation Plan	23
8.2	Materials Usage Agreement.....	23
8.3	Materials Identification and Usage List.....	23
8.4	Life Test Plan and Final Report for Lubricated Mechanisms.....	23
8.5	Additive Manufacturing Control Plan (AMCP)	23
8.6	AM Part Production Plan (PPP).....	23
9	CONTAMINATION CONTROL.....	24
9.1	Contamination Control Plan	24
9.2	Material Outgassing.....	24
9.3	Foreign Object Damage/Debris Program.....	24
10	METROLOGY AND CALIBRATION.....	25
10.1	Metrology and Calibration Program	25
10.2	Use of Calibrated and Non-Calibrated Instruments.....	25
11	GIDEP ALERTS AND PROBLEM ADVISORIES	26
11.1	Government-Industry Data Exchange Program.....	26
11.2	Alert Disposition.....	26
11.3	GIDEP Reporting.....	26
11.4	Review Reporting	26
12	END ITEM ACCEPTANCE DATA PACKAGE	27
	APPENDIX A DATA ITEM DESCRIPTIONS	28
	APPENDIX B ABBREVIATIONS AND ACRONYMS	68

List of Tables

Table 4-1. GSFC FMECA RPN Table	14
Table A-1 Data Item Descriptions	28

A. SCOPE, ORGANIZATION, AND CONVENTIONS

This document establishes a risk-based application of mission assurance requirements (MAR) for the Sounder for Microwave Based Applications (SMBA) instrument, a Class C instrument per National Aeronautics and Space Administration (NASA) Procedural Requirement (NPR) 8705.4. It is organized around the following safety and mission assurance (SMA) disciplines/focus areas, each with its own section:

Section 1 – General SMA Requirements	Section 7 – EEE Parts
Section 2 – Quality Management System	Section 8 – Materials and Processes
Section 3 – System Safety	Section 9 – Contamination Control
Section 4 – Reliability	Section 10 – Metrology and Calibration
Section 5 – Software Assurance	Section 11 – GIDEP Alerts and Problem Advisories
Section 6 – Workmanship	Section 12 – End Item Acceptance Data Package

Each discipline/focus area is further decomposed into subsections associated with its individual SMA activities or deliverables (e.g., Section 2.2, “Quality Management System,” subsection 2.2.2 flows down requirements for operating a Material Review Board (MRB)).

In this document:

1. A specific requirement, where contractor compliance is mandatory, is denoted by “**shall**,”
2. A good practice by “**should**,”
3. Permission for the contractor or Government is denoted by “**may**,”
4. An expectation by the Government is denoted by “**will**.”

The requirements in this document use the following identification (ID) convention:

SMA Discipline_Alphanumeric Identifier (e.g. QMS_01X).

Where the key is as follows: SMA Discipline	• GEN – General SMA Requirements • QMS – Quality Management System • SAF – System Safety • REL – Reliability • SWA – Software Assurance • WOR – Workmanship • EEE – EEE Parts • MAP – Materials and Processes • CON – Contamination Control • MAC – Metrology and Calibration • GID – GIDEP Alerts and Problem Advisories • EIA – End Item Acceptance Data Package
Numeric Identifier	A unique and sequential 2-digit number, with or without a character

Note: The contents of this MAR are the result of tailoring from a master set of requirements. Consequently, there may be instances where jumps or gaps occur in the numbering sequence of requirement IDs and data item descriptions (DID). Due to the tailoring, there may also be sections that

are not applicable, and have thus been marked “RESERVED.” This is done intentionally to preserve document organization and to facilitate tracking of data from implementation of requirements.

1 GENERAL

1.1 Systems Safety and Mission Assurance Program

GEN_02: The contractor **shall** submit a mission assurance compliance matrix that identifies variance(s) and acceptance rationale for processes, procedures, and standards that are proposed as alternatives to those specified by the contract (DID 1-1). This includes identification of items and MAR for which relief is requested via the Inherited Item Risk Assessment (IIRA) process (see Section 1.9, “Use of Inherited Products/Items”).

1.2 Management

GEN_03a: The contractor **shall** designate a manager for assurance activities.

GEN_03b: The assurance manager **shall** not be responsible for project costs and schedules other than those pertaining to assurance activities.

GEN_04: The contractor **shall** ensure that the assurance manager has direct access to upper management that is independent of project management and with functional freedom and authority to interact with all elements of the project.

1.3 Requirements Flow Down

GEN_05: The contractor **shall** ensure flow down of SMA requirements in this document to all suppliers based on the work to be performed and establish a process to verify compliance, with the exception of those approved for relief via the IIRA process (see Section 1.9, “Use of Inherited Products/Items”).

GEN_06: The contractor’s contract review and purchasing processes **shall** indicate the method for documenting, communicating, and reviewing requirements with sub-tier suppliers to ensure requirements are met.

GEN_07: The contractor **shall** ensure that quality plans, processes, procedures, hardware, and software submitted by the contractor’s sub-tier suppliers are compliant with the requirements in this IMAR, as applicable.

1.4 Identification of Project-Level Critical Items

GEN_08: The contractor **shall** identify its critical items for incorporation into the Government project-level critical items (PCI) with development in accordance with NPR 8735.2C, Section 4.1.4.

Identification of critical items and processes **should** include the results of system safety and reliability analyses.

1.5 Suspension of Work Activities

GEN_11: The contractor **shall** direct the suspension of any work activity that presents an unsafe work condition to personnel or imminent danger to property.

1.6 Surveillance

GEN_12: The contractor **shall** provide access to quality management system documentation, information systems and work products/artifacts to NASA representatives.

GEN_13: The work activities, operations, and documentation performed by the contractor and sub-tier contractors or suppliers **shall** be subject to evaluation, review, audit, inspection, and survey by NASA representatives at various points over the program/project development lifecycle.

GEN_14a: In accordance with Federal Acquisition Regulations (FAR) 46.103, 46.104, 46.202-2, 46.4, and 46.5, the contractor **shall** grant physical and/or remote access to NASA representatives to conduct an on-site and/or remote audit, assessment, or inspection upon notice. A no later than 30-day notice **will** be provided to the supplier/contractor prior to the start of an assessment.

GEN_14b: The contractor **shall** supply personnel, documents, records, equipment, and an acceptable work area within the contractor's facilities to assist with any audit, assessment, or inspection.

GEN_15: The contractor **shall** report the status of facility operations and quality metrics to NASA on a quarterly basis. This **should** include the following data for 1st tier and 2nd tier suppliers of PCIs:

- Quality escapes - Any product released by an internal or external supplier that is subsequently determined to be nonconforming to contract and/or product specification requirements.
- First pass yield - A measure of quality in a process that reflects the percentage of product made correctly without any rework or corrective activity.
- Supplier defect rate - The supplier defect rate measures the percentage of materials or product received from suppliers that do not meet required or compliance specifications.
- Internal audit results.

1.7 Government Mandatory Inspection Points

GEN_16: The contractor **shall** provide a proposed Government Mandatory Inspection Points (GMIP) plan, which will be used by the Government in determining the final GMIPs that **will** be conducted. NPR 8735.2 **may** be used as a guide.

GEN_19: Prior to the start of manufacturing, the contractor **shall** provide work instructions, procedures, drawings, etc., that are required for performing the planned inspections.

1.8 List of Sub-Tier Suppliers

GEN_20: The contractor **shall** provide a list of suppliers used for products produced under this contract (DID 1-2).

1.9 Use of Inherited Products/Items

Any products that will be build-to-print, rebuilt with modification, are commercial-off-the-shelf (COTS), or were previously developed and exist (e.g., spares) may be considered as candidate inherited items, subject to review by the Goddard Space Flight Center (GSFC) Standard Components (SC) Commodity Risk Assessment Engineer (-CRAE). With this process, the Government reviews the history of the product or similar variants, the experience of the manufacturer, and any design changes to determine whether the item can be accepted as an inherited item. If an item is accepted as an inherited item, the SC-CRAE will perform an IIRA to determine which SMA requirements are unnecessary, and make recommendations to the Chief Safety and Mission Assurance Officer (CSO). System Safety, Orbital Debris Assessment Report (ODAR), and End of Mission Plan (EOMP) requirements are not affected by the inherited items process.

1.10 Protection of Flight Hardware

GEN_22: The contractor **shall** evaluate the potential for Ground Support Equipment (GSE) to damage flight hardware and use appropriate means to prevent such damage from occurring.

GEN_23: The approach to obviate GSE damage to flight hardware **shall** be presented prior to the start of testing and at subsequent review milestones.

GEN_24: Prior to performing work on flight hardware, the contractor **shall** identify a list of items, if any, that are sensitive to normal handling environments, such as presence of light, presence of metal objects (magnets), or presence of humidity outside of normal cleanroom and workmanship limits.

GEN_25: The contractor **shall** hold a meeting on the day work is to be performed, prior to the start of each shift, that includes a discussion of the list of items.

1.11 Risk Management

SMA activities **should** be tightly linked with the project's Risk Management processes. For example, risks that evolve from reliability analyses that affect overall mission objectives **should** be managed in the project's risk database when not eliminated or mitigated to noncredible likelihood levels.

2 QUALITY MANAGEMENT SYSTEM

2.1 General

QMS_01: The contractor **shall** have a quality management system (QMS) that is compliant with the requirements of SAE AS9100, “Quality Systems - Aerospace - Model for Quality Assurance in Design, Development, Production, Installation, and Servicing.”

2.2 Supplemental Quality Management System Requirements

2.2.1 Control of Nonconforming Product

QMS_04a: The contractor **shall** have a documented closed loop system for identifying, reporting, and correcting product nonconformances.

QMS_04b: The system **shall** ensure that the adequacy of corrective action is determined by audit, inspection, or test, that objective evidence is collected, and that preventive action is implemented to preclude recurrence.

2.2.2 Material Review Board

QMS_06: The contractor **shall** have a documented process(es) for the establishment and operation of an MRB to process major non-conformances, which are those that affect form, fit, function, require a software change, involve the presence of foreign object debris (FOD), or those that the contractor determines involves elevated risk.

QMS_07: The contractor **shall** appoint a MRB chairperson who is responsible for implementing the MRB process and for appointing contractor representatives as MRB members.

QMS_09: The MRB process **shall** include a Government representative as a participating member on MRB actions involving major nonconformances.

QMS_11: The Government **shall** be provided notice and applicable documentation within 24 hours in advance of scheduled meetings for handling and dispositioning major non-conformances.

QMS_13: The MRB **shall** use the following disposition actions:

- Scrap — the product is not usable
- Re-work/Re-test — the product **will** be re-worked/re-tested to conform to requirements (sometimes referred to as “return to print”)
- Return to supplier — the product **will** be returned to the supplier
- Repair — the product **will** be repaired
- Use as is — the product **will** be used as is

2.2.3 Anomaly Reporting and Disposition

- QMS_14: The contractor **shall** have a documented process for the establishment and operation of an anomaly review board (ARB) to process (i.e., report and disposition) major anomalies, which are those that have resulted in hardware or software test failures and damage or potential damage to hardware.
- QMS_16: Reporting of major anomalies **shall** begin with the first application of power at the component level, flight software acceptance testing, when interfacing with flight hardware, and the first mechanical operation.
- QMS_19: The ARB (or equivalent function) **shall** permit a Government representative to be a participating member.
- QMS_20: The Government **shall** be provided notice within 24 hours of a major anomaly event. Applicable documentation in advance of scheduled ARB meetings.
- QMS_21: Anomalies that cannot be duplicated, have unknown root cause, or cannot be verified **shall** be assessed for residual risk, declared as red flag problem failure records, and brought to the Government's project risk board for disposition.

2.3 Orbital Debris Assessment Report and End of Mission Plan

- QMS_22: The contractor **shall** provide the information necessary for the development of the ODAR and the EOMP deliveries per the content defined in NASA-STD 8719.14 (DID 2-1).

3 SYSTEM SAFETY

3.1 General

SAF_01: The contractor **shall** document and implement a system safety program, support the Payload Safety Process as defined in chapter 3 of NPR 8715.7B, comply with launch service provider requirements, and comply with launch range safety requirements.

SAF_02a: The contractor's system safety program **shall** incorporate three independent inhibits in the design (dual failure tolerant) if a system failure may lead to a catastrophic hazard. A pre-launch catastrophic hazard is a payload-related hazard, condition, or event occurring prior to launch that could result in a fatal injury to personnel or loss of a ground facility. A post-launch catastrophic hazard is a payload-related hazard, condition, or event occurring after launch and up to payload separation that could result in a fatal injury or loss of flight termination system.

SAF_02b: The contractor's system safety program **shall** incorporate two independent inhibits in the design (single failure tolerant) if a system failure may lead to a critical hazard. A critical hazard is defined as a hazard, condition, or event that may cause severe injury or occupational illness or major property damage to facilities.

SAF_02c: The contractor's system safety program **shall** adhere to specific detailed safety requirements in this document, including compliance verification that must be met for design elements with hazards that cannot be controlled by failure tolerance. The process by which safety is incorporated into these design elements (e.g., structures and pressure vessels) is called "Design for Minimum Risk".

3.2 Mission Related Safety Requirements Documentation

SAF_03a: The contractor **shall** implement the launch range safety requirements that are applicable to the launch site.

SAF_03b: The contractor **shall** implement the most stringent safety requirement in the event there are conflicting requirements. The sections below include the applicable requirements documents for the pertinent launch ranges:

ELV Eastern Test Range or Western Test Range Missions

- NASA-STD 8719.24 (with Annex)
- Kennedy NPR (KNPR) 8715.3 applicable at Kennedy Space Center (KSC) property, KSC-controlled property, and offsite facility areas where KSC has operational responsibility.
- NPR 8715.7
- Launch Site Facility-specific Safety Requirements, as applicable (e.g., Astrotech)

Wallops Flight Facility (WFF) Missions

- NASA-STD 8719.24 (with Annex)
- GSFC-STD-8009

3.3 System Safety Deliverables

3.3.1 System Safety Plan

SAF_10: The contractor **shall** prepare a System Safety Plan (SSP) that describes the tasks and activities of system safety management and engineering required to identify, evaluate, and eliminate or control hazards to the hardware, software, and system design by reducing the associated risk to an acceptable level throughout the system life-cycle, including launch range safety requirements (DID 31).

3.3.2 Safety Requirements Compliance Checklist

SAF_11: The contractor **shall** document and implement a Safety Requirements Compliance Checklist to demonstrate that the payload complies with NASA and range safety requirements (DID 3-2).

3.3.3 Instrument Safety Assessment Report

SAF_12: The contractor **shall** generate an Instrument Safety Assessment Report (ISAR) to document the comprehensive evaluation of the risk being assumed prior to the testing or operation of an instrument. The spacecraft contractor **will** use the ISAR (DID 3-4ISR) as an input to the Safety Data Package (SDP).

3.3.4 Hazard Analyses

3.3.4.1 Preliminary Hazard Analysis

SAF_15: The contractor **shall** perform a Preliminary Hazard Analysis (PHA) to obtain an initial risk assessment and to identify safety critical areas of a concept or system. The PHA is based on the best available data, including mishap data from similar systems and other lessons learned.

SAF_16a: The contractor **shall** evaluate hazards associated with the proposed design or function for severity, control approach (fault tolerance or design for minimum risk), and operational constraints.

SAF_16b: The contractor **shall** identify safety provisions and alternatives that are needed to eliminate hazards or reduce their associated risk to an acceptable level.

SAF_17: The contractor **shall** deliver the PHA with the preliminary ISAR (DID 3-4ISR).

3.3.4.2 Operations Hazard Analysis and Hazard Verification Tracking Log

- SAF_18: The contractor **shall** document, implement, and maintain an Operations Hazard Analysis and a hazard Verification Tracking Log (VTL) to demonstrate that hardware operations, test equipment operations, and integration and test (I&T) activities comply with the safety requirements of the facilities where the activities **will** be performed and that hazards associated with those activities are mitigated to an acceptable level of risk (DID 3-3).
- SAF_19: The contractor **shall** update and maintain the hazard VTL during I&T activities to track open issues.

3.3.4.3 Operating and Support Hazard Analysis

- SAF_20: The contractor **shall** perform an Operating and Support Hazard Analysis (O&SHA) to evaluate activities for hazards introduced during testing, transportation, storage, integration, and pre-launch operations at the launch site. The primary purpose is to evaluate the adequacy of procedures used to eliminate, control, or mitigate identified hazards to ensure implementation of safety requirements for personnel, procedures, and equipment during activities at the launch site.
- SAF_21: The contractor **shall** submit the results of the O&SHA as a part of the intermediate & final ISARs (DID 3-4ISR).

3.3.5 Verification Tracking Log

- SAF_22: The contractor **shall** document and implement a VTL that documents a Hazard Control and Verification Tracking process as a closed-loop system that ensures safety compliance has been satisfied per applicable launch range safety requirements.
- SAF_23: The contractor **shall** document in the VTL the process of verifying the control of hazards by test, analysis, inspection, similarity to previously qualified hardware, or any combination of these activities.
- SAF_24: The contractor **shall** ensure that verifications listed on the hazard reports refer to specific test, analysis, or inspection reports with a summary of the pertinent results.
- SAF_25: The contractor **shall** make the results of these tests, analyses, and inspections available for Government review.
- SAF_26a: The VTL **shall** identify hazard controls that are not verified as closed.
- SAF_26b: The VTL **shall** be delivered with the final ISAR (DID 3-4ISR).
- SAF_27: The contractor **shall** provide regular electronic updates of the VTL until all hazard controls are verified as closed.

3.3.6 Hazardous Procedures for Payload I&T and Pre-launch Processing

SAF_28: The contractor **shall** document the hazardous procedures that **will** be implemented when I&T and pre-launch activities are performed at processing facilities and the launch site (DID 3-5).

SAF_29: The contractor **shall** ensure that the procedures comply with applicable facility safety requirements.

SAF_30: The contractor **shall** provide safety support for the implementation of hazardous procedures.

3.3.6.1 Lifting Device Safety Requirements

SAF_31: The contractor **shall** implement the following safety requirements for lifting devices and equipment (LDE) when performing NASA work at non-NASA facilities:

- Overhead cranes, winches, and wire rope hoists **shall** have a dual hoist braking system installed per the NASA STD 8719.9C (Lifting Standard). Chain hoists typically do not have dual brakes as standard equipment unless included in the manufacturer's design when specifically requested by the user. A single hoist motor holding brake in combination with a Variable Frequency Drive dynamic braking system is an acceptable dual braking system.
- Label and tag LDE, per NASA STD 8719.9C, Section 4.9, requirements, with no more than its Working Load Limit (WLL) as determined by the original equipment manufacturer (OEM) or its current certified WLL if that value is lower than the OEM rated WLL.
- Perform an initial one-time proof load test per the following NASA STD 8719.9C, Section 4.5, requirements:
 - 1.25X WLL for overhead cranes.
 - 1X WLL for mobile elevated work platforms that are used near critical hardware.
 - 1X WLL for mobile cranes and derricks (.95X to 1X is acceptable).
 - 1.25X WLL for below-the-hook lifting devices.
 - Slings (i.e., wire rope, synthetics, chain, etc.) **shall** only be proof load tested beyond its WLL with OEM approval.
 - 2X WLL for rigging hardware items used for critical lifts (i.e., shackles, hoist rings, turnbuckles, etc.).
Perform a 1X WLL load test every four years after the initial proof test on all LDE.
 - In addition to visual inspection, perform a post-load test non-destructive test inspection (e.g., radiographic, ultra-sonic, magnetic particle, dye penetrant, etc.) on crane hooks and critical welds. A critical weld is one in which a failure would result in a failure of the hardware. The inspections **shall** be performed by an American Society of Non-destructive Testing (ASNT) or equivalently trained inspector.

3.3.7 Mishap Reporting and Investigation

SAF_32: The contractor **shall** prepare a Pre-Mishap Plan that describes appropriate mishap and close call notification, reporting, recording, and investigation procedures (DID 3-6).

SAF_33: The contractor **shall** report accidents, test failures, or other mishaps and close calls promptly to NASA.

SAF_34: The contractor **shall** promptly investigate test failures, or other mishaps and close calls to determine the root cause.

3.3.8 NASA Payload Safety Program Forms

SAF_35: The contractor **shall** prepare NASA Payload Safety Program Forms. The forms are available at <https://kscsma.ksc.nasa.gov/PayloadSafety/forms>

4 RELIABILITY

4.1 Reliability Program Plan

REL_01: The contractor **shall** document and implement a Reliability Program Plan (RPP) that includes both qualitative and quantitative techniques to support decisions regarding mission success and safety throughout system development (DID 4-1).

REL_03: The contractor **shall** include a detailed approach to the analysis of hardware and software for their contributions to system reliability and mission success.

The contractor **should** perform reliability analyses concurrent with design and maintain consistency between reliability analyses so that identified problem areas are addressed, and corrective action taken in a timely manner.

4.2 Analysis of Design

4.2.1 Failure Modes and Effects Criticality Analyses and Critical Items List

REL_11a: The contractor **shall** perform, document, and maintain a Failure Modes and Effects Criticality Analysis (FMECA) that addresses flight hardware and software and GSE that interfaces with flight systems that are being designed, built, or provided from project initiation through launch and mission operations.

REL_11b: The contractor's FMECA **shall** include likelihood, cause, detection and mitigation, and the effects of each failure mode at the local, subsystem, and system or mission levels.

REL_11c: Components and systems designated as inherited items **shall** be analyzed at the interface level, all others at the box or functional level (DID 4-2).

REL_12: The contractor **shall** prepare and maintain a Critical Items List (CIL) for items with failure-mode severity categories 1SC, 1, 2S, and 2, per Table 4-1.

Table 4-1. GSFC FMECA RPN Table

Criticality Rating	Technical		Failure Type	Failure Severity	Detection/Prevention (Mitigations)
	Likelihood of Occurrence	Consequence			
5 Very High	$0.50 < P_F$	Minimum mission success criteria are not achievable.	Critical and SPF	1SC Failures that could cause a catastrophic event such as the loss of life, permanently disabling injury to personnel, or facility loss/destruction.	None - Failure will not be detected and will not be prevented or mitigated.
				1 Failures (and failures with failure recoveries) that could result in mission loss or minimum mission success criteria not being achievable.	
4 High	$0.25 < P_F \leq 0.50$	Major impact to full mission success criteria. Minimum mission success criteria are achievable.	Critical	2S Failures that could prevent detection, mitigation, or operations during a hazardous condition resulting in 1SC conditions, eliminate a hazard inhibit, or cause severe injury or occupational illness or major property damage.	Remote - Unlikely failure will be detected or prevented or mitigated.
				2 Failures (and failures with failure recoveries) that could result in loss of one or more mission objectives, including the substantial loss of data, loss of functionality, or reduction in life of the mission, but minimum mission success criteria are still achievable.	
3 Moderate	$0.15 < P_F \leq 0.25$	Moderate impact to full mission success criteria. Minimum mission success criteria are achievable with margin.	Significant	3 Failures (and failures with failure recoveries) that could result in loss of one mission objective, including the significant loss of data, loss of functionality, or reduction in life of the mission, but minimum mission success criteria are still achievable.	Low to Moderate - Failure may be detected and may be prevented or mitigated.
2 Low	$0.02 < P_F \leq 0.15$	Minor impact to higher priority full mission success criteria.	Minor	4 Significant failures that could cause degradation to full mission objectives and still meet minimum mission.	Moderate to High - Failure is likely to be detected before occurrence and has a good chance of being prevented or mitigated.
				4T Failures that cause a temporary 1, 2S, or 2 consequence condition until an identical provision or equivalent provisions are used to resolve with no more than degradation to mission success.	
1 Very Low	$0.001 < P_F \leq 0.02$	Minor impact to lower priority full mission success criteria.		5 Minor failures that could result in insignificant or no loss to mission objectives.	Certain - failure will be detected and prevented or mitigated.
0 Very Very Low	$P_F \leq 0.001$	N/A – Highly Implausible or Noncredible			

- REL_13: The contractor **shall** prepare and maintain a Single Point Failure (SPF) list for modes resulting in category 1 and 1SC severities, per Table 4.-1. and document applicable failure causes, mitigations, and retention rationale.
- REL_14: The contractor **shall** identify and assess any known common cause failure modes and their causes for redundant components where loss of that component set would result in severity categories 1SC, 1, 2 and 2S. Such information should be provided as notes/remarks on the FMECA spreadsheet.
- REL_15: The contractor **shall** estimate the likelihood score for each failure mode from 0-5 using the appropriate criteria from table 4-1 and Goddard Procedural Requirement (GPR) 7120.4, as necessary, or another scale approved by GSFC Reliability Engineering. Each likelihood estimation can be based on qualitative assessment and/or failure rate data from other analyses (i.e., system calculations) in order to score each failure mode for the mission duration.

REL_16: The contractor **shall** identify the consequence for each failure mode using the appropriate criteria from table 4-1 and GPR 7120.4, as necessary.

4.2.2 Fault Tree Analysis

REL_17: The contractor **shall** perform and maintain qualitative Fault Tree Analyses (FTAs), as deemed necessary by the CSO and the Mission Systems Engineer (MSE) (DID 4-3).

REL_18: The fault tree **shall** address both hardware and software contributions at a level necessary to identify risks, verify mitigations, and assist in the development of fault management.

REL_19: In the event the contractor or the project identifies a major mission risk contributor in the FMECA or FTA, the contractor **shall** quantify (and if necessary, expand) the appropriate FTA or the portions of the FTA necessary for detailed risk assessment, as deemed necessary by the CSO and the MSE.

FTA **should** also be considered for use to check the FMECAs for completeness.

4.2.3 Reliability Calculations

REL_21: The contractor **shall** perform and maintain reliability and ground-system availability calculations (DID 4-4) using FTA, reliability block diagrams, and/or Probabilistic Risk Assessment (PRA) to identify design weaknesses, support design trades, and demonstrate the impact of critical items, as deemed necessary by the CSO and the MSE.

4.3 Limited Life Analysis

REL_22: The contractor **shall** perform a Limited Life Item Analysis (LLA) that includes expected life, required life, an assessment of life margin (including servicing and maintenance); provides retention rationale for items with an expected life of less than 2x the required life and fosters a plan to manage limited life items (DID 4-5).

REL_25: The risk assessment and mitigations plans **shall** factor in wear caused by atomic oxygen, solar and trapped radiation, shelf-life, extreme temperatures, thermal cycling, write/erase cycles and mechanical wear or fatigue, and include refurbishment and maintenance plans.

4.4 Parts Stress Analysis

- REL_26: The contractor **shall** perform parts stress and derating analyses for electrical, electronic, and electromechanical (EEE) parts in accordance with GSFC EEE-INST-002, or the contractor's own established standard, unless specifically relieved of this requirement by the GSFC IIRA process (See Section 1.9) (DID 4-6).
- REL_28: If alternate derating guidelines are requested to be used, they **shall** be submitted to the Parts Control Board and GSFC Reliability Engineering for approval.

4.5 Worst-Case Analysis

- REL_30: The contractor **shall** perform worst-case analyses (WCA) for circuit designs that are new, or significantly modified, or are being used in an environment not covered by the existing analysis and identified as critical through analysis of the design (See Section 4.2) (DID 4-7).

5 SOFTWARE ASSURANCE

5.1 Software Assurance Program

SWA_01: The contractor **shall** establish a software assurance program consisting of a planned and systematic set of activities and disciplines that ensures that software conforms to organizational and project-specific requirements and standards throughout the project lifecycle, where software is defined as:

- Computer programs, procedures, and possibly associated documentation and data pertaining to the operation of a computer system.
- All or a part of the programs, procedures, rules, and associated documentation of an information processing system.
- Program or set of programs used to run a computer.
- All or part of the programs which process or support the processing of digital information.
- Part of a product that is the computer program or the set of computer programs.

Note: The software definition applies to software developed by NASA, software developed for NASA, software maintained by or for NASA, COTS software, Government-off-the-shelf (GOTS), modified-off-the-shelf software (MOTS), open-source software, reused software components, auto-generated code, embedded software, software used on GSE, the software executed on processors embedded in programmable logic devices, legacy, heritage, applications, freeware, shareware, trial or demonstration software, and open-source software components.

SWA_03: The contractor **shall** ensure the independence of software assurance from software engineering.

SWA_04: The contractor **shall** document and implement a Software Assurance Plan and Schedule compliant to NASA-STD-8739.8B (DID 5-1).

The plan **should** include the software assurance processes, procedures, tools, and techniques to be used commensurate with the software classification and software safety criticality assessment, with additional tailoring in accordance with guidance provided by NASA-STD-8739.8B for each category of software (new, reused, off-the-shelf, auto-generated code, etc.).

The plan addresses both Software Assurance and Software Safety disciplines. This **should** include the necessary collaboration with relevant SMA and Engineering stakeholders (i.e., system safety, system reliability, hardware quality, system security, and software engineering), and the process by which traceability is established to their respective analyses and/or requirements.

5.2 Surveillance of Software Development, Maintenance, and Assurance Activities

SWA_08: Consistent with the general requirement for support of Government surveillance (see Section 1.6, “Surveillance”), the contractor **shall** provide on-request access to following:

- Software problem reports
- Software documentation (i.e., management plans, assurance plans, configuration management plans, requirements specifications, design documents, test plans, test cases, test procedures, test results, software review results, software engineering and assurance schedule, maintenance plans)
- Source code
- Findings and corrective actions from software process and product audits and assessments

6 WORKMANSHIP

6.1 General

WOR_02: The contractor **shall** implement a workmanship program to assure that electronic packaging technologies, processes, and workmanship meet mission objectives for quality and reliability per the requirements of the following standards:

- NASA-STD-8739.6 Implementation Requirements for NASA Workmanship Standards, excluding sections 8.1, 9.1, and 10.1.
- J-STD-001
- GSFC-STD-6001
- IPC-2225
- IPC-6015

WOR_04: The contractor **shall** comply with one of the following standards for electrical cables and harnesses:

- NASA-STD-8739.4
- IPC/WHMA-A-620-S

6.2 Electrostatic Discharge Control

WOR_06: The contractor **shall** prepare and implement an electrostatic discharge (ESD) control plan that conforms to the requirements of ANSI/ESD S20.20 (DID 6-1).

6.3 Printed Circuit Boards

WOR_08: The contractor **shall** comply with one of the following standards for rigid printed circuit boards (PCB):

- IPC-6012 (Note: the latest revision preferred, but older revisions **may** be acceptable based on inherited designs or contractor standard practices)
- MIL-PRF-55110H
- ECSS-Q-ST-70-60

Note: In cases in which high-density interconnect components (e.g., reconfigurable field programmable gate arrays (FPGA), high-density static random-access memory , etc.) or board population are present, the contractor **should** consult with the Project and Government Microelectronic Packaging and Circuit Board (MPCB) CRAE for a specification strategy to avoid extensive rebuilds or the incorporation of risky features solely in order to meet the specification. This consultation **may** result in deviations from the specifications above with no waiver required.

WOR_10: The contractor **shall** document and implement a PCB procurement plan (DID 6-2).

WOR_12: For non-conforming coupons, the evaluation report **shall** be delivered to the GSFC MPCB CRAE for a risk assessment. If the MPCB CRAE's risk assessment

determines that there is no elevated risk, no waiver is required and the CSO will determine the disposition of the board.

6.4 Lead-Free Control Measures

WOR_16: The contractor **shall** document and implement a Lead-Free Control Plan (LFCP) (DID 6-5).

WOR_17: The contractors **shall** submit uses of lead-free solder or surface finishes to the NASA project office for approval before use.

7 ELECTRICAL, ELECTRONIC, AND ELECTROMECHANICAL (EEE) PARTS

7.1 General

EEE_02: The contractor **shall** document and implement a Parts Control Plan (PCP) per Level 3 instructions of GSFC EEE-INST-002 (DID 7-1).

Note: The contractor **may** use, as is, Class V, S, Q, B, M compliant microcircuits; Joint Army Navy Space (JANS), JANTXV, and JANTX semiconductors without any further approval. The contractor **may** use, as is, automotive or high-reliability COTS parts, compliant to Level 3, per NASA-STD-8739.10, without any additional screening or qualification tests and without any further approval.

7.2 Parts Control Board

EEE_07: The contractor **shall** establish a Parts Control Board (PCB) that is responsible for the planning, management, and coordination of the selection, application, and procurement requirements of EEE parts.

EEE_10: The contractor **shall** identify the person responsible for directing and managing the EEE parts program and interfacing with Government assurance personnel.

EEE_13: The contractor **shall** include the GSFC Parts Engineer or the Parts and Radiation Assurance Engineer as a participating member of the PCB.

7.3 Re-use of EEE Parts

EEE_15: The contractor **shall** require approval of the PCB to re-use EEE parts that have been installed.

7.4 Master EEE Parts List

EEE_16: The contractor **shall** develop and deliver a Master EEE Parts List and maintain it for the duration of the project (DID 7-2).

7.5 Radiation Effects Mitigation

EEE_19: The contractor shall provide a plan, for approval, for how radiation will be addressed (DID 7-3).

8 MATERIALS AND PROCESSES

8.1 M&P Selection, Control, and Implementation Plan

MAP_01a The contractor **shall** prepare and implement a Materials and Processes (M&P) Selection, Control, and Implementation Plan (MPSCIP) (DID 8-1).

MAP_01b: NASA-STD-6016 **shall** be used as a baseline, with contractor standard practices acceptable if associated command media are shared with NASA.

8.2 Materials Usage Agreement

MAP_03: The contractor **shall** prepare Materials Usage Agreements (MUA) (DID 8-2).

8.3 Materials Identification and Usage List

MAP_04: The contractor **shall** prepare a Materials Identification and Usage List (MIUL) (DID 8-3).

8.4 Life Test Plan and Final Report for Lubricated Mechanisms

MAP_05: The contractor **shall** prepare and implement a life test plan and final report for lubricated mechanisms (DID 8-4).

8.5 Additive Manufacturing Control Plan (AMCP)

MAP_06: The contractor **shall** prepare and implement an Additive Manufacturing Control Plan (AMCP) for the design and manufacture of Additively Manufactured (AM) parts (DID 8-5).

8.6 AM Part Production Plan (PPP)

MAP_07: The contractor **shall** prepare a Part Production Plan (PPP) for each AM part (DID 8-6).

9 CONTAMINATION CONTROL

9.1 Contamination Control Plan

CON_01: The contractor **shall** prepare and implement a contamination control program (DID 9-1).

9.2 Material Outgassing

CON_03: The contractor **shall** include in DID 8-3 information regarding material outgassing.

CON_04: If an alternate standard to NASA-STD-6016 is implemented, materials **shall** meet requirements of < 1% total mass loss (TML) and < 0.1% collected volatile condensable material (CVCM) at 125°C under vacuum for twenty-four hours when tested to ASTM E595, “Standard Test Methods for Total Mass Loss and Collected Volatile Condensable Materials from Outgassing in a Vacuum Environment”.

9.3 Foreign Object Damage/Debris Program

CON_05: The contractor **shall** prepare and implement a foreign object damage/debris (FOD) program (DID 9-2).

10 METROLOGY AND CALIBRATION

10.1 Metrology and Calibration Program

MAC_01: For measurement and test equipment that has documented requirements for metrological accuracy and traceability, the contractor **shall** comply with NASA-STD-8739.12 or verify equipment against calibrated instruments or intrinsic standards, using a documented procedure.

Note: The contractor **may** verify torque wrenches against a calibrated torque tester prior to use.

10.2 Use of Calibrated and Non-Calibrated Instruments

MAC_02: The contractor **shall** record the measurements that require accuracy in applicable project build documents (e.g., work order authorizations, job orders, task sheets, or test plans), including the article of calibrated equipment used to take the measurement and its calibration end date.

MAC_03: When verification is chosen instead of calibration, the contractor **shall** perform verification within a timeframe that has been demonstrated to provide appropriate levels of reliability, in the same facility, and under the same conditions that **will** be encountered during the process.

11 GIDEP ALERTS AND PROBLEM ADVISORIES

11.1 Government-Industry Data Exchange Program

GID_01: The contractor **shall** participate in Government-Industry Data Exchange Program (GIDEP) per the GIDEP Operations Manual (Note: this document is available through <http://www.gidep.org>).

11.2 Alert Disposition

GID_03: The contractor **shall** review the following, hereafter referred to collectively as Alerts, for effects on EEE parts, materials, equipment, and software used in NASA products:

- GIDEP Alerts
- GIDEP SAFE-ALERTS
- GIDEP Problem Advisories
- GIDEP Agency Action Notices
- NASA Advisories

GID_05: When the contractor identifies an item in their design, inventory, or assembly that is documented in an Alert, the contractor **shall** disposition the item and Alert through the MRB as a major nonconformance.

11.3 GIDEP Reporting

GID_06: The contractor **shall** prepare and submit failure experience data and safety issue reports per the requirements of GIDEP Operations Manual whenever failed or nonconforming items are discovered that are available to other buyers.

11.4 Review Reporting

GID_07: The contractor **shall** report the status of NASA products that are affected by Alerts (or by significant EEE parts, materials, software, and safety problems) and the actions taken to eliminate or mitigate negative effects at monthly status reviews, parts control board meetings, program milestone reviews, and readiness reviews.

12 END ITEM ACCEPTANCE DATA PACKAGE

EIA_01: The contractor **shall** maintain the End Item Acceptance Data Package (EIADP) throughout the project lifecycle and submit in accordance with (DID 12-1).

Appendix A Data Item Descriptions

This Appendix provides mission assurance DID's referenced in the IMAR. The delivery due dates and category (e.g. for approval) for these items are contained in NS1-SMBA-CDRL-0001, which can be accessed from https://ipdtdms.gsfc.nasa.gov/frontmenu_dsp.cfm. The associated Contract Data Requirements List (CDRL) item is listed alongside the IMAR DID number within each DID.

Table A-1 Data Item Descriptions

DID #	Title
1-1	Mission Assurance Compliance Matrix
1-2	Supplier List
2-1	Orbital Debris Assessment Report (ODAR) and End of Mission Plan (EOMP)
3-1	System Safety Plan
3-2	Safety Requirements Compliance Checklist
3-3	Operations Hazard Analysis and Hazard Verification Tracking Log
3-4ISR	Instrument Safety Assessment Report
3-5	Hazardous Procedures for Payload I&T and Pre-Launch Processing
3-6	Pre-Mishap Plan
4-1	Reliability Program Plan
4-2	FMECA and Critical Items List
4-3	Fault Tree Analysis
4-4	Reliability Calculations
4-5	Limited Life Items List
4-6	Parts Stress Analysis
4-7	Worst-Case Analysis
5-1	Software Assurance Plan
6-1	ESD Control Plan
6-2	Printed Circuit Board Procurement Plan
6-5	Lead-Free Control Plan
7-1	EEE Parts Control Plan
7-2	Master EEE Parts List
7-3	Radiation Effects Mitigation
8-1	M&P Selection, Control, and Implementation Plan (MPSCIP)
8-2	Materials Usage Agreement (MUA)
8-3	Materials Identification and Usage List (MIUL)
8-4	Life Test Plan and Final Report for Lubricated Mechanisms
8-5	Additive Manufacturing Control Plan (AMCP)
8-6	AM Part Production Plan (PPP)
9-1	Contamination Control Plan
9-2	Foreign Object Debris Program
12-1	End Item Acceptance Data Package

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-1/1-1

2. Title:

Mission Assurance Compliance Matrix

3. Reference:

N/A

4. Use:

The Mission Assurance Compliance Matrix shall address the contractual system safety and mission assurance requirements.

5. Preparation Information:

The Mission Assurance Compliance Matrix shall address the contractual system safety and mission assurance requirements as applied to:

- All flight hardware and software that is designed, built, or provided by the contractor and its subcontractors, or furnished by the Government, from project initiation through launch and mission operations.
- The ground support equipment that interfaces with flight equipment to the extent necessary to assure the integrity and safety of flight items.
- The ground data system, if applicable.
- The Contractor shall indicate whether it and its subcontractors will comply, not comply or partially comply. In a comment column, the Contractor shall indicate the entity (Contractor or name of subcontractor) that will not comply, and for partial non-compliance describe the extent of the non-compliance. The Contractor shall also indicate whether it seeks relief for the nonconformance under IMAR Section 1.9.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-2/1-2

2. Title:

Supplier List

3. Reference:

N/A

4. Use:

Provide a list of suppliers used for products produced under this contract.

5. Preparation Information:

Provide the following information for suppliers that have been contracted to provide products (down to the component-level):

- a) Supplier name
- b) Location(s)
- c) Cage code(s)
- d) Product description
- e) Contract start and end dates (if available)
- f) Delivery date (if available)

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-3/2-1

2. Title:

Orbital Debris Assessment Report (ODAR) End of Mission Plan (EOMP)

3. Reference:

1. NASA-STD-8719.14 Process for Limiting Orbital Debris
-

4. Use:

Provide information for development of the ODAR and the EOMP deliveries per the content defined in NASA-STD 8719.14

5. Preparation Information:

NASA-STD-8719.14 Process for Limiting Orbital Debris Appendix A (ODAR) and Appendix B (EOMP) provide details on what information is required for the Project office to complete these analyses.

NOTE: Orbital Debris Assessment Software is available for download from Johnson Space Center at URL: <http://sn-callisto.jsc.nasa.gov/mitigate/das/das.html>.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-4/3-1

2. Title:

System Safety Plan

3. Reference:

1. NASA-STD-8719.24 (with Annex), NASA Launch Vehicle Payload Safety Requirements

4. Use:

The contractor shall prepare a SSP that describes the development and implementation of a system safety program that complies with the requirements of NPR 8715.7, the launch service provider, and launch range safety.

5. Preparation Information:

The contractor shall

- a) Define the roles and responsibilities of personnel.
- b) Define the required documentation, applicable requirements documents, and completion schedules for analyses, reviews, and safety packages.
- c) Address support for Safety Reviews, Safety Working Group Meetings and Technical Interchange Meetings.
- d) Provide for early identification and control of hazards to personnel, facilities, support equipment, and the flight system during product development, including design, fabrication, test, transportation, and ground activities.
- e) Address compliance with the launch range safety requirements.
- f) Include a safety review process that meets the requirements of NPR 8715.7 Payloads Safety Program.
- g) Address compliance with safety requirements imposed by NASA and Occupational Safety and Health Administration (OSHA), address design and operational safety requirements, and address contractually imposed mission unique obligations.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-5/3-2

2. Title:

Safety Requirements Compliance Checklist

3. Reference:

1. NASA-STD 8719.24 (with Annex), NASA Payload Safety Requirements

4. Use:

The contractor shall prepare a compliance checklist of all design, test, analysis, and data submittal requirements.

5. Preparation Information:

The following shall be included:

- a) Criteria and requirement.
- b) System
- c) Indication of compliance, noncompliance, or not applicable
- d) Rationale for indications other than compliant
- e) Resolution
- f) Reference
- g) Copies of Range Safety and NASA approved non-compliances, including waivers and equivalent levels of safety certifications.

Note: the contractor shall submit safety waivers for non-compliant design elements per paragraph 3.3.7.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-6/3-3

2. Title:Operations Hazard Analysis and Hazard
Verification Tracking Log3. Reference:

N/A

4. Use:

To demonstrate hazards associated with hardware operations, test equipment operations, and integration and test (I&T) activities are mitigated to an acceptable level of risk.

5. Preparation Information:

The OHA shall include the following information:

- a) Introduction – a summary of the major findings of the analysis and the proposed corrective actions and definitions of special terms, acronyms, and abbreviations; the findings will address the safety requirements of the facilities in which activities will be performed.
- b) System Description – a description of system hardware and configuration, with a list of subsystem components and schedules for integration and testing.
- c) Analysis of Hazards.
- d) List of real or potential hazards to personnel, equipment, and property during I&T processing.
- e) The following information shall be included for each hazard:
 - o System Component/Phase – the phase and component with which the analysis is concerned, e.g., system, subsystem, component, operating/maintenance procedure, or environmental condition.
 - o System Description and Hazard Identification, Indication:
 - o A description of expected results from operating the component/subsystem or performing the operating/maintenance action.

-
- o A complete description of the actual or potential hazard resulting from normal actions or equipment failures; indicate whether the hazard will cause personnel injury and equipment damage.
 - o A description of warning indicators for the operator/crew that includes all means of identifying the hazard to operational/maintenance personnel.
 - o A description of the safety hazards of software controlling hardware systems where the hardware effects are safety critical.
 - o Effect on System – the detrimental effects of an uncontrolled hazard on the system
 - o Risk Assessment.
 - o Caution and Warning Notes – a list of warnings, cautions, procedures required in operating and maintenance manuals, training courses, and test plans.
 - o Status/Remarks – the status of actions to implement hazard controls.
 - o References (e.g., test reports, preliminary operating and maintenance manuals, and other hazard analyses)

Note: OHA controls for engineering test units undergoing environmental tests shall be presented in accordance with local safety authority requirements

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-7/3-4ISR

2. Title:

Instrument Safety Assessment Report

3. Reference:

N/A

4. Use:

The ISAR will identify safety features of the hardware, software, and system design as well as procedural, hardware, and software related hazards that may be present in the instrument. This includes specific procedural controls and precautions that should be followed.

5. Preparation Information:

The ISAR will include the following information:

- a) The safety criteria and methodology used to classify and rank hazards, including assumptions upon which the criteria or methodologies were based or derived.
- b) The results of hazard analyses and tests used to identify hazards in the system including:
- c) Those hazards that still have a residual risk and the actions that have been taken to reduce the associated risk to a level contractually specified as acceptable.
- d) Results of tests conducted to validate safety criteria, requirements, and analyses.
- e) Hazard reports documenting the results of the hazard analyses to include a list of all significant hazards along with specific safety recommendations or precautions required to ensure safety of personnel, property, or the environment. NOTE: Identify whether or not the risks may be expected under normal or abnormal operating conditions.

The contractor will submit with the Final ISAR a signed statement that:

- f) Hazards have been closed out through elimination or through risk mitigations that are contractually specified as acceptable.
- g) Identifies hazards that have not been eliminated or controlled at the time of submission.
- a) The instrument is ready to test, operate, or proceed to the next phase.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-8/3-5

2. Title:Hazardous Procedures for Payload I&T and Pre
-Launch Processing3. Reference:

1. NASA-STD 8719.24 (with Annex), NASA Payload Safety Requirements
2. KNPR 8715.3 KSC Safety Practices Procedural Requirements (as applicable)

4. Use:

Documents the hazardous procedures and associated safeguards that will be used for integration and test activities and pre-launch activities.

5. Preparation Information:

The safeguards will comply with the applicable safety requirements for the installation where the activities will be performed.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-9/3-6

2. Title:

Pre-Mishap Plan

3. Reference:

1. Sample Pre-Mishap Plan – available upon request.
-

4. Use:

The Pre-Mishap Plan shall identify the processes and procedures to be followed to respond to the occurrence of a mishap or a close call and identify the chain of individuals, including government personnel, to be contacted.

5. Preparation Information:

. The Pre-Mishap Plan includes the following information:

a. The contractor's policies and plan regarding response to a mishap or close call, to include:

1. Actions to be taken from the occurrence through implementation of corrective actions.
2. Plans for emergency response, notification, evidence preservation, mishap investigation, the mishap investigation report, lessons learned, and corrective actions.
3. Information regarding responsible for duties and tasks involved in the process.

b. The following definitions:

1. Close Call -- An occurrence or a condition of employee concern in which there is no injury or minor injury requiring first aid and no or minor equipment or property damage (less than \$20,000) but which possesses a potential to cause a mishap.
2. Incident -- An occurrence of a close call or a mishap.
3. Mishap -- An unplanned occurrence that results in damage to property or personnel injury or illness: damage to contractor, Government, or customer-owned hardware property or critical products; fatalities, injuries, or illnesses occurring during program operations; environmental releases or spills occurring in the course of program operations.

c. The following definitions regarding the type of mishaps:

1. Type A Mishap -- A mishap resulting in one or more of the following: (1) an occupational injury or illness resulting in a fatality, a permanent total disability, or the hospitalization for inpatient care of 3 or more people within 30 workdays of the mishap; (2) a total direct cost of mission failure and property damage of \$2 million or more.

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2. Type B Mishap -- A mishap that caused an occupational injury or illness that resulted in a permanent partial disability, the hospitalization for inpatient care of 1-2 people within 30 workdays of the mishap, or a total direct cost of mission failure and property damage of at least \$500,000 but less than \$2,000,000.
 3. Type C Mishap -- A mishap resulting in a nonfatal occupational injury or illness that caused any days away from work, restricted duty, or transfer to another job beyond the day or shift on which it occurred, or a total direct cost of mission failure and property damage of at least \$50,000 but less than \$500,000.
 4. Type D Mishap -- A mishap that caused any nonfatal OSHA recordable occupational injury and/or illness that does not meet the definition of a Type C mishap, or a total direct cost of mission failure and property damage of at least \$20,000 but less than \$50,000.
- d. Contact information for Project Office personnel.
 - e. Notification schedule and mishap response process timeline (notification in no more than 24 hours).
 - f. Note: The following are not reportable as mishaps but may be reportable as failures or anomalies:
 1. Property Damage:
 2. Items normally covered under Failure Reporting
 3. Malfunction or failure of component parts or equipment due to normal wear and tear where the malfunction is the only damage, and the only action is to replace or repair the equipment.
 4. Anticipated damage to equipment or property was incurred during testing or manufacturing.
 5. Property damage from vandalism, arson, sabotage or acts of God.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-10/4-1

2. Title:

Reliability Program Plan

3. Reference:

1. NPD 8720.1, NASA Reliability and Maintainability (R&M) Program Policy
2. NASA-STD-8729.1, Planning, Developing and Managing an Effective Reliability and Maintainability (R&M) Program.
3. NPR 8705.4 Risk Classification for NASA Payloads.

4. Use:

The Reliability Program Plan shall explain how the contractor intends to implement and comply with MAR Reliability analysis requirements.

5. Preparation Information:

The explanation shall include the following:

- a) The organizational responsibilities and functions for each task to be performed as part of the program.
- b) The point of contact for implementing and generating the necessary documents for each requirement.
- c) A list of subsystems and components for which trend analysis will be performed, including the parameters to be monitored.
- d) The execution and management plan for each task, including schedule.
- e) Documentation, methods, scope, level of detail, and reporting specific to each task in the plan and for each element if there are differences.
- f) Alternate/additional methodologies (non-analysis, e.g., qual testing, modeling, inheritance assessment) for supporting analyses and addressing requirements.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-11/4-2

2. Title:

FMECA and Critical Items List

3. Reference:

1. NPR 8705.4 Risk Classification for NASA Payloads
2. GPR 7120.4 Risk Management

4. Use:

The FMECA Report shall explain the approach of the analysis, methodologies, assumptions, results, conclusions, and recommendations.

5. Preparation Information:

The report shall include the following:

- a) Objectives.
- b) Level of the analysis.
- c) Ground rules.
- d) Functional description.
- e) Functional block diagrams.
- f) Reliability block diagrams to illustrate any redundancies.
- g) Equipment analyzed.
- h) Data sources used.
- i) Problems identified.
- j) Corrective actions
- k) Work sheets identifying item functions, failure modes; causes; severity category; consequence category; effects at the item, next higher level, and mission level; detection methods; and mitigating provisions.
- l) Critical Items List (CIL) for severity categories 1SC, 1, 1R, 1S, and 2, including item identification, cross-reference to FMECA line items, and retention rationale. Appropriate retention rationale may include design features, historical performance, acceptance testing, manufacturing product assurance, corrective action recommendation/elimination of undesirable failure modes, proper design controls, and failure detection methods. The rationale also will contain data that describes operational constraints caused by occurrence of the failure and any measures that can be taken to restore the function on orbit where known.
- m) SPF list for failure modes resulting in category 1 and 1SC severities per table 4-1.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-12/4-3

2. Title:

Fault Tree Analysis (FTA)

3. Reference:

1. NASA Fault Tree Handbook with Aerospace Applications
(<http://www.hq.nasa.gov/office/codeq/doctree/fthb.pdf>)
2. NPR 8705.4 Risk Classification for NASA Payloads
3. NPR 8715.3 NASA General Safety Program Requirements

4. Use:

The qualitative Fault Tree Analyses (FTAs) is a top-down approach to identify system faults.

5. Preparation Information:

The mission FTA Reports shall contain:

- a) Analysis ground rules including definitions of undesirable end states.
- b) References to documents and data used.
- c) Fault tree diagrams.
- d) Results and conclusions.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-13/4-4

2. Title:

Reliability Calculations

3. Reference:

1. NASA/SP-2009-569 Bayesian Inference for NASA Probabilistic Risk and Reliability Analysis
2. NASA Fault Tree Handbook with Aerospace Applications

4. Use:

To identify design weaknesses, support design trades, and demonstrate the impact of critical items.

5. Preparation Information:

The reliability/availability calculations reports shall include the following:

- a) The methodology and results of comparative reliability assessments including mathematical models
- b) Reliability block diagrams and/or fault tree diagrams
- c) Failure rates
- d) Mean time to repair (ground systems)
- e) Maintenance man hours/hours (ground systems)
- f) Failure definitions / definitions of undesirable end states
- g) References to documents and data used
- h) Trade-offs considered
- i) Assumptions
- j) Any other pertinent information used in the assessment process
- k) Results and conclusions

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.: 2. Title:

MA-14/4-5

Limited Life Items List

3. Reference:

N/A

4. Use:

Provides expected life, required life, and an assessment of life margin (including servicing and maintenance); provides retention rationale for items with an expected life of less than 2x the required life and fosters a plan to manage limited life items.

5. Preparation Information:

The contractor shall prepare and maintain a list of limited life items and their predicted impact on mission operations. The list shall include expected life, required life, duty cycle, and rationale for selecting and using the items for which the expected life is less than twice the required life. See sample content and format below.

Item	Life-Limiting Mechanism	Expected Life	Required Life	Compliance & Life Ratio	Data Sources & Notes
Mirror Coating	Degradation of optical properties	10 years	5 years	Complies 2.0	Expected Life: Vendor datasheet Required Life: Requirement Doc 98765
Switch	Wear-out of contacts	190 cycles	100 cycles	Does Not Comply (DNC) 1.9	Expected Life: Test Report 12345 Required Life: Requirement Doc 98765

Example Limited Life Items List

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-15/4-6

2. Title:

Parts Stress Analysis

3. Reference:

1. GSFC EEE-INST-002 Instruction for EEE Parts Selection, Screening, Qualification, and Derating
-

4. Use:

Provides data on EEE part stresses and derating plans.

5. Preparation Information:

The Parts Stress Analysis Report shall contain analysis worksheets that include:

- a) Identification of derating standard used.
- b) Part identification (traceable to circuit diagrams).
- c) Rated stress.
- d) Applied stress (consider all significant operating parameter stresses at the extremes of anticipated environments).
- e) Ratio of applied-to-rated stress.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-16/4-7

2. Title:

Worst-Case Analysis

3. Reference:

N/A

4. Use:

Provides data for circuit designs that are new or significantly modified or are being used in an environment not covered by the existing analysis and identified as critical through analysis of the design.

5. Preparation Information:

The WCA report shall include the following:

- a) Identification of worst-case conditions considered for each component.
- b) Consideration of critical parameters at maximum and minimum limits.
- c) The effect of environmental stresses on the operational parameters being evaluated.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-17/5-1

2. Title:

Software Assurance Plan

3. Reference:

1. NASA-STD-8739.8, NASA Software Assurance and Software Safety Standard
2. NPR 7150.2, NASA Software Engineering Requirements

4. Use:

Provides plan for software assurance using NASA-STD-8739.8B.

5. Preparation Information:

The SAP shall address the following:

- a) Purpose and Scope
- b) Assurance Activities, Tools, and Techniques by discipline:
 1. Software Quality (process and product)
 2. Software Safety
 3. Software Reliability
 4. Software Verification and Validation
 5. Independent Verification and Validation (if applicable)
- c) Software Assurance Program Metrics
- d) Problem Reporting and Corrective Action
- e) Assurance records, collection, maintenance, and retention
- f) Risk Management
- g) NASA-STD-8739.8 Compliance Matrix

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-18/6-1

2. Title:

ESD Control Plan

3. Reference:

1. ANSI/ESD S20.20-2014, Protection of Electrical and Electronic Parts, Assemblies And Equipment (Excluding Electrically Initiated Explosive Devices)
-

4. Use:

Provides plan for ESD control using ANSI/ESD S20.20.

5. Preparation Information:

The ESD Control Plan shall be prepared and implemented to comply with ANSI/ESD S20.20 requirements and the ESD sensitivity of the product being developed.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-19/6-2

2. Title:

Printed Circuit Board Procurement Plan

3. Reference:

1. IPC-2221 Generic Standard on Printed Board Design
2. IPC-601X

4. Use:

Provides plan for procurement of printed circuit boards.

5. Preparation Information:

For all printed boards to be used in mission hardware, the procurement plan shall contain:

- a) Name of next higher-level assembly (box or subsystem).
- b) Printed board assembly name.
- c) Part number.
- d) Indication as being a new or heritage design.
- e) Description of design complexity, to include:
 - 1) Number of layers.
 - 2) Overall board thickness.
 - 3) Most complex via stack-up.
 - 4) Use of micro-vias.
 - 5) Use of via-in-pad.
 - 6) Line width and spacing.
 - 7) Maximum voltage in application.
 - 8) Base material (IPC-4101 designation or trade name).
 - 9) Applicable printed board design standard.

-
- 10) Applicable printed board performance standard.
 - 11) Fabrication notes or procurement specifications.
 - 12) Note of any waiver or deviation affecting printed board requirements or acceptability.
 - 13) Printed board supplier(s) or candidates.
 - 14) Quantity required for flight build.
 - 15) Minimum quantity required for spares.

The procurement information will ensure that test coupons are fabricated for each design in a quantity sufficient to meet testing requirements per IPC-2221, Generic Standard on Printed Board Design and satisfy supplier acceptance testing per IPC-601X.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-20/6-5

2. Title:

Lead-Free Control Plan

3. Reference:

N/A

4. Use:

Provides plan for lead-free solder or surface finish control.

5. Preparation Information:

The LFCP shall meet the requirements of GEIA-STD-0005-1 and GEIA-STD-0005-2 for tin-based solders and surface finishes that are less than 3% lead by weight and comply with the Level 2C requirements set.

Note: The LFCP requires prior MRB approval for uses of tin-based lead-free solder or surface finishes and whisker mitigation methods.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-21/7-1

2. Title:

EEE Parts Control Plan

3. Reference:

1. GSFC EEE-INST-002 Instructions for EEE Parts Selection, Screening, Qualification, and Derating
2. S-311-M-70 Specification for Destructive Physical Analysis
3. AS5553 Counterfeit Electronic Parts; Avoidance, Detection, Mitigation, and Disposition

4. Use:

To document a Parts Control Plan (PCP) per Level 3 instructions of GSFC EEE-INST-002 Instruction for EEE Parts Selection, Screening, Qualification, and De-rating.

5. Preparation Information:

The PCP shall address the following:

- a) Parts control program organization and management
- b) EEE Parts control per GSFC EEE-INST-002 Instructions for EEE Parts Selection, Screening, Qualification, and Derating
- c) Shelf-life control plan
- d) Supplier and manufacturer surveillance
- e) Procedures regarding application specific integrated circuits, gate arrays, system-on-chip, and custom integrated circuits
- f) Incoming inspection and test
- g) Sparing policies
- h) Destructive physical analysis per S-311-M-70 Specification for Destructive Physical Analysis
- i) Defective parts controls program.
- j) Handling, preservation, and packing
- k) Contamination control

- l) Alternate quality conformance inspection and small lot sampling
- m) Traceability and lot control
- n) Failure analysis
- o) Counterfeit parts control plan per AS5553 Counterfeit Electronic Parts, Avoidance, Detection, Mitigation, and Disposition
- p) Radiation hardness assurance program, which shall address total ionizing dose; displacement damage (total non-ionizing dose); destructive and non-destructive single-event effects; single-event effect rates; proton hardness/tolerance
- q) Parts Control Board Operations
- r) Organization and membership
- s) Meeting schedule and notices
- t) Distribution of meeting agenda, notes, and minutes
- u) Review and approval responsibilities and processes
- v) Documentation and records

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-22/7-2

2. Title:

Master EEE Parts List

3. Reference:

N/A

4. Use:

The contractor shall maintain the Master EEE Parts List Information in a searchable electronic format. The GSFC Project Parts Engineer shall have access to the list.

5. Preparation Information:

The contractor shall generate and maintain a Master Parts List with the minimum information listed below for the various stages throughout the project's lifecycle:

a) For EEE parts that are identified for potential flight use, the list shall include the following information:

1. Flight component identity to the circuit board level
2. Complete part number (i.e., Defense Supply Center Columbus part number, Specification Control Drawing part number, with all suffixes)
3. Manufacturer's Generic Part number
4. Manufacturer (not distributor)
5. Part Description (please include meaningful detail)
6. Federal Supply Class
7. Procurement Specification
8. Comments and clarifications, as appropriate
9. Estimated quantity required (for procurement forecasting)

b) For EEE parts that are approved for flight use the list shall include the following information:

-
1. Procurement Part Number
 2. Flight Part Number (if different from the procurement part number)
 3. Package Style/Designation
 4. PCB Status
 5. PCB Approval Date
 6. PCB Required Testing/Evaluations

c) For EEE parts that are approved for as-designed use the list shall include the following information:

1. Assembly Name/Number
2. Next Level of Assembly
3. Need Quantity
4. Reference Designator(s)
5. Item number (if applicable)

d) For EEE parts in flight hardware the list shall include the following information:

1. Assembly serial number
2. Item revision
3. Next Level of Assembly serial number
4. Lot/Date/Batch/Heat/Manufacturing Code, as applicable
5. Manufacturer's Cage Code (specific plant location when relevant)
6. Distributor/supplier, if applicable
7. Part number
8. Part serial number (if applicable)

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-23/7-3

2. Title:

Radiation Effects Mitigation

3. Reference:

N/A

4. Use:

Provide the overall plan for mitigating the effects of radiation.

5. Preparation Information:

The radiation effects mitigation plan provides one or more of the following approaches, as applicable.

1. Avoidance: dormancy of sensitive electronic elements in high stress regions such as South Atlantic Anomaly (SAA) or Van Allen Belts.
2. Radiation-Hardened-By-Design RHBD: Proven rad-hard by design approach, applied to circuits and/or parts.
3. Traditional parts-centric: Use of Radiation Hardness Assurance parts with radiation-tolerant design to accommodate high stress region operation.
4. Modern parts-centric: Use of familiar sensitive parts¹ along with proven circuit designs in comparable environment.
5. Radiation-tolerant design: Use radiation-tolerant circuit design techniques including features such as Metal-Oxide Semiconductor Field-Effect Transistor (MOSFET) protection and overcurrent detection with reset capability, resettable processors, Error Detection and Correction (EDAC), derating beyond EEE-INST-002 recommendations, etc.
6. Risk-based approach combining past on-orbit experiences in similar stressing environments.
7. System fault-tolerance (including redundancy): This may include new, unproven approaches, with backup proven systems.

¹Sensitive parts include memory, processors, Complementary Metal-Oxide Semiconductor (CMOS) devices, MOSFETs, etc.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-24/8-1

2. Title:M&P Selection, Control, and Implementation Plan
(MPSCIP)3. Reference:

1. NASA-STD-6016C Standard Materials and Processes Requirements for Spacecraft
2. NASA-STD-8739.14 NASA Fastener Procurement, Receiving Inspection, and Storage Practices for NASA Mission Hardware
3. 541-PG-8072.1.2 Goddard Space Flight Center (GSFC) Fastener Integrity Requirements

4. Use:

Provides the plan for materials and processes.

5. Preparation Information:

For each Material and Process Requirement (MPR) in NASA-STD-6016C, Section 4, the contractor shall prepare and implement an MPSCIP that identifies the degree of conformance under the subheading “Degree of Conformance,” and identifies the method of implementation under the subheading “Method of Implementation.” In addition to the MPRs, the MPSCIP shall include the contractor fastener management and control policy per NASA-STD-8739.14 or 541-PG-8072.1.2, including degree of conformance.

The use of well-established contractor practices is encouraged to meet the intent of MPRs, and may be referenced by the MPSCIP, provided documentation of these practices is made available for review remotely by the GSFC MPE (in accordance with electronic access requirements specified in Section 1.6, “Surveillance”). Tailoring of NASA-STD-6016C requirements by the MPSCIP is permitted, pending review and approval by GSFC. Upon approval, the MPSCIP becomes the M&P implementation document used for verification (see NASA-STD-6016C, Section 4.1.1)

Notes:

- For NASA-STD-6016C, Section 4.1.5., the contractor may use GSFC forms or any equivalent electronic format in lieu of the MAPTIS format with approval from the GSFC MPE.
- The contractor may use the GSFC outgassing database (URL <http://outgassing.nasa.gov>) in addition to MAPTIS (URL <http://maptis.nasa.gov>).

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-25/8-2

2. Title:

Materials Usage Agreement (MUA)

3. Reference:

1. NASA-STD-6016C Standard Materials and Processes Requirements for Spacecraft
2. MSFC-STD-3029A Guidelines for The Selection Of Metallic Materials For Stress Corrosion Cracking Resistance In Sodium Chloride Environment

4. Use:

To provide information to demonstrate a material application is technically acceptable, but does not meet NASA-STD-6016C.

5. Preparation Information:

The MUA documentation approach, per NASA-STD-6016C, Section 4.1.6, shall be defined in the MPSCIP (see DID 8-1).

The MUA package shall include the technical information required to justify the application. MUAs for stress corrosion shall include a Stress Corrosion Cracking Evaluation Form per MSFC-STD-3029A (see NASA-STD-6016C) and a stress analysis.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-26/8-3

2. Title:

Materials Identification and Usage List (MIUL)

3. Reference:

N/A

4. Use:

Provide a material list for materials used on the instrument.

5. Preparation Information:

The MIUL documentation approach, per NASA-STD-6016C, Section 4.1.5, shall be defined in the MPSCIP (DID 8-1).

Additions and Prescribed Changes to NASA-STD-6016C MIUL requirements:

- a) The MIUL shall identify all unique materials and processes used in the construction of the deliverable hardware.
- b) With approval from the GSFC MPE, for NASA-STD-6016C, Section 4.1.5, the contractor may use GSFC forms or the contractor's equivalent forms in lieu of the MAPTIS format.
- c) The MIUL shall include Thermal Vacuum Stability data, when appropriate, per NASA-STD-6016C, Section 4.2.3.6.
- d) The MIUL shall include the vacuum bake-out schedule for materials that fail outgassing requirements or provide a separate MUA (see DID 8-2).
- e) The contractor may use the GSFC outgassing database (URL <http://outgassing.nasa.gov>) in addition to MAPTIS (URL <http://maptis.nasa.gov>).
- f) The MIUL shall document, when appropriate, compliance to MSFC-STD-3029A
- g) The MIUL shall clearly identify all materials that do not meet the requirements of NASA-STD-6016C and require an MUA per Section 8.2 and DID-8-2 of this MAR.
- h) The MIUL shall include soldering flux and solvents used for cleaning flight electronic assemblies, other than isopropyl alcohol or deionized water.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-27/8-4

2. Title:

Life Test Plan and Final Report for Lubricated Mechanisms

3. Reference:

1. NASA-STD-6016C Standard Materials and Processes Requirement for Spacecraft
 2. NASA-TM-86556 Lubrication Handbook for the Space Industry (Part A: Solid Lubricants, Part B: Liquid Lubricants)
 3. NASA/CR-2005-213424 Lubrication for Space Applications
-

4. Use:

The contractor shall prepare and implement a Life Test Plan to qualify all lubricated mechanisms either by life testing or heritage with an identical mechanism used in an identical application. The contractor shall perform a lubricant loss analysis for all mechanisms to show that the design meets a 10X margin.

5. Preparation Information:

The plan shall contain:

- a. Table of Contents
- b. Description of lubricated mechanisms, performance functions, summary of subsystem specification, and life requirements.
- c. Heritage of identical mechanisms and descriptions of identical applications.
- d. Design, drawings, and lubrication system used by the mechanism.
- e. Test plan, including vacuum, temperature, and vibration test environmental conditions.
- f. Criteria for a successful test.

NASA-STD-6016C, Section 4.2.3.4, NASA-TM-86556, and NASA/CR-2005-213424 should be used as guides when developing the plan.

Alternatively, the contractor shall provide detailed justification for NOT conducting life tests and/or lubricant loss analyses.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-28/8-5

2. Title:

Additive Manufacturing Control Plan (AMCP)

3. Reference:

1. NASA-STD-6030 Additive Manufacturing Requirements for Spaceflight Systems
 2. NASA-STD-6016C Standard Materials and Processes Requirement for Spacecraft
-

4. Use:

The contractor shall prepare and implement an AMCP that states the AMR from NASA-STD-6016C, identifies the degree of conformance under the subheading “Degree of Conformance,” and identifies the method of implementation under the subheading “Method of Implementation.”

5. Preparation Information:

For each Additive Manufacturing Requirement (AMR) in NASA-STD-6030, Sections 4 through 8, with the additions and prescribed changes below,

The use of well-established contractor practices is encouraged to meet the intent of AMRs, and may be referenced by the AMCP, provided documentation of these practices is made available for review remotely by the GSFC MPE (in accordance with electronic access requirements specified in Section 1.5, “Surveillance”). Tailoring of NASA-STD-6030 requirements by the AMCP is permitted, pending review and approval by GSFC. Upon approval, the AMCP becomes the AM implementation document used for verification (see Section 4.2.1.c of NASA-STD-6030)

Additions and Prescribed Changes to NASA-STD-6030:

- Technologies not included in NASA-STD-6030, Section 1.4.2, Table 1, will be considered pending review and approval by the GSFC MPE.
- A Mission Critical Part is specified by the project-defined critical items list.
- Exempt Parts, per NASA-STD-6030, Section 4.3.1.4, are not allowed.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-29/8-6

2. Title:

AM Part Production Plan (PPP)

3. Reference:

1. NASA-STD-6030 Additive Manufacturing Requirements for Spaceflight Systems
-

4. Use:

The primary purpose of the PPP for Class C parts is to document the rationale for the Class C designation.

5. Preparation Information:

The PPP shall document the part-specific information required by NASA-STD-6030, Section 7.

Note: Not all AMRs in NASA-STD-6030, Section 7, apply to Class C parts.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-30/9-1

2. Title:

Contamination Control Plan

3. Reference:

1. GSFC-STD-7000 General Environmental Verification Standard (GEVS)
2. GSFC-STD-1000 Rules for the Design, Development, Verification, and Operation of Flight Systems
3. ASTM E595 Standard Test Methods for Total Mass Loss and Collected Volatile Condensable Materials from Outgassing in a Vacuum Environment
4. Outgassing Data for Selecting Spacecraft Materials (URL: <http://outgassing.nasa.gov/>)

4. Use:

To provide a plan for contamination control to prevent degradation to materials or design features.

5. Preparation Information:

The contractor shall provide material properties data; design features; test data; system tolerance of degraded performance; methods to prevent degradation. The items below shall be addressed in the plan:

- a) Beginning of life and end of life requirements for contamination sensitive surfaces or subsystems
- b) Methods and procedures used to measure and maintain the levels of cleanliness required during each of the various phases of the item's lifetime (e.g., protective covers, environmental constraints, purges, cleaning/monitoring procedures)
- c) Materials.
- d) Outgassing as a function of temperature and time.
- e) Nature of outgassing chemistry.
- f) Areas, weight, location, view factors of critical surfaces.
- g) Venting: size, location and relation to external surfaces.
- h) Thermal vacuum test contamination monitoring plan, to include vacuum test data, quartz crystal microbalance (QCM) location and temperature, pressure data, system temperature profile, and shroud temperature.
- i) On-orbit spacecraft and instrument performance as affected by contamination deposits.
- j) Contamination effect monitor.

-
- k) Methods to prevent and recover from contamination in orbit.
 - l) Evaluation of on-orbit degradation.
 - m) Photopolymerization of outgassing products on critical surfaces.
 - n) Atomic oxygen erosion and re-deposition.
 - o) Analysis of contamination impact on the satellite on orbit performance.
 - p) In orbit contamination impact from other sources such as space station and adjacent instruments.
 - q) Ground/Test support equipment controls to prevent contamination of flight item(s).
 - r) Facility controls and processes to maintain hardware integrity (protection and avoidance).
 - s) Transportation/shipping controls and processes to maintain hardware integrity (protection and avoidance).
 - t) Training.
 - u) Data package on test results for materials and as-built product.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-31/9-2

2. Title:

Foreign Object Debris Program

3. Reference:

1. AS9146 Foreign Object Damage (FOD) Prevention Program – Requirements for Aviation, Space, and Defense Organizations

4. Use:

Provides a plan for FOD prevention.

5. Preparation Information:

The plan will address the preservation of product with respect to foreign object debris prevention per the requirements of AS9146 Foreign Object Damage (FOD) Prevention Program - Requirements for Aviation, Space, and Defense Organizations.

DATA ITEM DESCRIPTION

1. CDRL No./IMAR DID No.:

MA-32/12-1

2. Title:

End Item Acceptance Data Package

3. Reference:

N/A

4. Use:

Provides a data package to include all paperwork necessary to assess the quality of the end item.

5. Preparation Information:

The contractor prepares the EIADP as part of design development and implementation such that it is completed prior to delivery. The following items shall be included (as specified by requirement):

- a) The deliverable item name, serial number, part number, and classification status (e.g., flight, non-flight, ground support, etc.).
- b) Appropriate approval signatures (e.g., contractor's quality representative, product design lead, Government Representative, etc.).
- c) List of shortages or open items at the time of acceptance with supporting rationale.
- d) As-built serialization.
- e) As-built configuration.
- f) In-process Work Orders (available for review at contractors--not a deliverable).
- g) Final assembly and test Work Order.
- h) Major MRB actions.
- i) Major anomaly reports.
- j) Acceptance testing procedures and report(s), including environmental testing.
- k) Trend data
- l) Anomaly/problem failure reports with root cause and corrective action dispositions.

- m) Master EEE parts list.
- n) As-built materials list.
- o) Chronological history, including:
- p) Total operating hours and failure-free hours of operation.
- q) Total number of mechanical cycles and remaining cycle life.
- r) Limited life items, including data regarding the life used and remaining.
- s) As-built final assembly drawings.
- t) PCB coupon results.
- u) Photographic documentation of hardware (pre- and post-conformal coating for printed wiring assemblies, box or unit, subsystem, system, harness, structure, etc.).
- v) Waivers
- w) Certificate of Compliance which is signed by management.

Appendix B Abbreviations and Acronyms

ARB	Anomaly Review Board
AM	Additively Manufactured
AMCP	Additive Manufacturing Control Plan
AMR	Additive Manufacturing Requirement
ANSI	American National Standards Institute
ASNT	American Society for Nondestructive Testing
CCB	Configuration Control Board
CDR	Critical Design Review
CDRL	Contract Data Requirements List
CIL	Critical Items List
COTS	Commercial off-the-shelf software
CSO	Chief Safety and Mission Assurance Officer
CVCM	Collected Volatile Condensable Material
CRAE	Commodities Risk Assessment Engineer
DID	Data Item Description
EEE	Electrical, Electronic, and Electromechanical
EIADP	End Item Acceptance Data Package
EOMP	End of Mission Plan
ESD	Electrostatic Discharge Control
ETC	Estimate To Complete
FAR	Federal Acquisition Requirements
FMEA	Failure Modes and Effects Analysis
FMECA	Failure Modes and Effects Criticality Analysis
FOD	Foreign Object Debris
FSM	Funded Schedule Margin
FTA	Fault Tree Analysis
GIDEP	Government-Industry Data Exchange Program
GMIP	Government Mandatory Inspection Point
GOTS	Government off-the-shelf software
GPR	Goddard Procedural Requirement
GSFC	Goddard Space Flight Center
GSE	Ground Support Equipment
I&T	Integration and Test
IIRA	Inherited Item Risk Assessment
IMAR	Instrument Mission Assurance Requirements
ISAR	Instrument Safety Assessment Report
KNPR	Kennedy NASA Procedural Requirement
KSC	Kennedy Space Center
LDE	Lifting Devices and Equipment
LEO	Low Earth Orbit
LFCP	Lead-Free Control Plan
LV	Launch Vehicle

LLA	Limited Life item Analysis
M&P	Materials and Processes
MAR	Mission Assurance Requirements
MAPTIS	Materials and Processes Technical Information System
MOSFET	Metal-Oxide Semiconductor Field-Effect Transistor
MOTS	Modified off-the-shelf
MPCB	Materials Process Control Board
MPSCIP	Materials and Processes Selection, Control, and Implementation Plan
MRB	Material Review Board
MSE	Mission Systems Engineer
MUA	Materials Usage Agreement
MIUL	Materials Identification and Usage List
NASA	National Aeronautics and Space Administration
NEON	Near Earth Orbit Network
NPD	NASA Policy Directive
NPR	NASA Procedural Requirement
NS1	Near Earth Orbit Network Project Series 1
O&SHA	Operating and Support Hazard Analyses
ODAR	Orbital Debris Assessment Report
OEM	Original Equipment Manufacturer
OHA	Operations Hazard Analysis
OSHA	Occupational Safety and Health Administration
PCI	Project Level Critical Items
PCB	Printed Circuit Board
PCB	Parts control board
PCP	Parts Control Plan
PDR	Preliminary Design Review
PHA	Preliminary Hazard Analyses
PPP	Part Production Plan
PRA	Probabilistic Risk Assessment
PSR	Pre-Ship Review
RPP	Reliability Program Plan
SAE	Society of Automotive Engineers
SAP	Software Assurance Plan
SC	Standard Components
SDP	Safety Data Package
SMA	Safety and Mission Assurance
SMBA	Sounder for Microwave-based Applications
SPF	Single Point Failure
SRR	System Readiness Review
SSPP	System Safety Program Plan
STD	Standard
TBD	To Be Determined
TBR	To Be Resolved
TBS	To Be Scheduled
TML	Total Mass Loss

VTL	Verification Tracking Log
WCA	Worst Case Analysis
WFF	Wallops Flight Facility
WLL	Working Load Limit